

Zero-Token Re-Run Architecture in CADABRA

How to Save, Cache, and Re-Execute CAD Scripts in SOLIDWORKS Without Token Consumption

01 Core Principle: How Zero-Token Re-Execution Works

When you run an initial prompt in CADABRA, the cloud LLM converts your request into native SOLIDWORKS API commands (C# or VBA). Token consumption occurs **only during this LLM generation step**. Once generated, the execution is purely local via SOLIDWORKS COM interop.

- **First Run (Token Consuming):** User Prompt → Cloud LLM Processing (Token Cost) → Generates API Code → Executes in SOLIDWORKS.
- **Re-Run (Zero Token):** Local Script / Macro File (.swp) → Local COM API Execution → Re-builds Model in SOLIDWORKS (**0 Tokens, 100% Offline**).

02 3 Ways to Re-Run Operations Without Burning Tokens

- **Method A: Native Macro Export (.swp)**
In CADABRA Task Pane, click "Save as Macro". This saves a local .swp file. To re-run, go to **Tools > Macro > Run** in SOLIDWORKS or map it to a hotkey.
- **Method B: CADABRA Local Favorites Caching**
CADABRA stores previously generated API execution payloads in a local SQLite cache. Clicking "Re-run from History" executes the stored code directly on the COM bus without sending tokens to the cloud.
- **Method C: SOLIDWORKS CommandManager Toolbar Button**
Assign the exported .swp macro directly to a custom button on your SOLIDWORKS CommandManager for instant single-click execution.

03 Comparison: Macro Re-Run vs. Live Agent Prompting

Metric / Feature	Saved Macro Re-Run (.swp)	Live CADABRA Agent Call
Token Cost	0 Tokens (Free)	Consumes tokens per prompt
Execution Speed	Instant (< 100ms)	1.5s – 5.0s (Cloud latency)
Internet Requirement	Offline (Local COM API)	Online Cloud Connection
Adaptability	Static (Exact programmed steps)	Dynamic (Adapts to new models)

04 Step-by-Step Implementation Guide

1. **Step 1:** Prompt CADABRA to generate your complex geometry, hole pattern, or drawing automation.
2. **Step 2:** In the CADABRA Task Pane panel, click "Export Macro" and choose your local directory.
3. **Step 3:** Open SOLIDWORKS, navigate to **Customize > Commands > Macro**.
4. **Step 4:** Drag a custom Macro Button onto your SOLIDWORKS CommandManager bar and point it to your saved .swp file.
5. **Step 5:** Click the button anytime you need the operation. The task executes in milliseconds with zero token usage.